

Allowing any unsigned integral type as parameter type for literal operators

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Introduction

User-defined literals are a great tool for shortening code that would otherwise need to be overly verbose, but since the first proposal, they lacked one feature that would make writing code even simpler - supporting arbitrary unsigned types as arguments for `operator""`. Consider the following code:

```
constexpr std::uint8_t operator "" _foo(unsigned long long value)
{
    return value;
}

int main()
{
    auto foo = 1024_foo;    // this compiles with no warnings, even though it overflows, whereas this:
    char bar = 1024;        // warns about changing the value
}
```

It should be obvious that the first should also produce a diagnostic, since it might be a result of a simple programming mistake. There is also another problem, which appears to be far less important, but still exists. Consider the following platform specific code for a platform, where `std::uint64_t` is the biggest unsigned integer type:

```
constexpr std::uint64_t operator "" _bar(std::uint64_t value)
{
    // ...
}
```

results in an error in both GCC and Clang, because they use `unsigned long`, not `unsigned long long` for `std::uint64_t`, at least on Linux. This is an environment-specific problem, but the Standard should provide a way to overcome it in a portable way.

Proposed solution

The proposed solution is as follows:

1. Allow any unsigned type as an argument for `operator ""`, not `unsigned long long` only.
2. Make narrowing in case of user defined literals ill-formed.

An additional feature gained with 1) and 2) is allowing to limit the range of values used in a user-defined literal.

Note: this proposal doesn't propose `operator ""` overloading for unsigned integral types. Allowing such overloading would create ambiguities; it is not desirable.

Proposed wording changes

Change in 13.5.8 `over.literal` paragraph 3:

The declaration of a literal operator shall have a *parameter-declaration-clause* equivalent to one of the following:

- `const char*`
- `unsigned char`
- `unsigned short int`
- `unsigned int`
- `unsigned long int`
- `unsigned long long int`
- **an extended unsigned integer type ([basic.fundamental].3)**
- ...

Add two new paragraphs before 13.5.8 `over.literal` paragraph 5:

Literal operator overloading for unsigned integral types is not supported. If there exist two or more literal

operator declarations with the same literal suffix identifier with a single parameter, but of different unsigned integral types, the program is ill-formed.

If a narrowing conversion would happen when a literal operator with a single parameter of an unsigned integral type is invoked implicitly through user-defined literal, the program is ill-formed.

Change in 13.5.8 over.literal paragraph 8:

```
template <char...> int operator "" _j(const char*); // error: invalid parameter-declaration-clause
std::uint8_t operator "" _k(std::uint8_t);
auto l = 256_k; // error: integer constant is too large for its type
std::uint16_t operator "" _k(std::uint16_t); // error: literal operator overloading is not supported
```